BEING LOCALLY OF FINITE TYPE IS FPPF LOCAL ON THE SOURCE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The general criterion for a property of morphisms to be fppf local on the source. Same chapter, Lemma lemma-properties-morphisms-local-source, invoked at proof lines 7059-7062. Reduces the claim to four clauses, of which the first three are elementary and the fourth is the descent statement.
- Zariski locality, preservation under composition, and the inclusion of flat locally finitely presented in locally finite type. Morphisms, Lemmas morphisms-lemma-locally-finite-type-characterize, morphisms-lemma-composition-finite-type and morphisms-lemma-finite-presentation-finite-type, invoked at proof lines 7052-7058. Supplies clauses (1), (2) and (3).
- Finite type descends along a surjective etale map of affine schemes. Same chapter, Lemma lemma-finite-type-local-source-fppf-algebra, invoked at proof lines 7061-7062. Supplies clause (4).

Source reference. Original lemma 0367: sources/stacks-descent/pinned-original-source.tex, source lines 4569–4569. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 036H: sources/stacks-descent/pinned-original-source.tex, source lines 6920–6920. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01T3: sources/stacks-chow/tags.txt, source lines 2360–2360. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

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Source reference. Original lemma 01TW: sources/stacks-chow/tags.txt, source lines 2389–2389. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01T2: sources/stacks-chow/tags.txt, source lines 2359–2359. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *The property $\mathcal{P}(f) =$"f is locally of finite type" is fppf local on the source.*

Proof. Being locally of finite type is Zariski local on the source and the target, see Morphisms, Lemma 01T2. It is a property which is preserved under composition, see Morphisms, Lemma 01T3, and a flat morphism locally of finite presentation is locally of finite type, see Morphisms, Lemma 01TW. This proves (1), (2) and (3) of Lemma 036H. The final condition (4) is Lemma 0367. Hence we win. □